Muhammad Achad Ansori

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Robotics Engineer specializing in embedded systems, control systems, and computer vision. Experience spans marine robotics, offshore operations, smart manufacturing, aquaculture technology, and demolition robotics, from embedded programming and PCB design to autonomous navigation and AI inspection on edge devices. Works across both hardware and software, turning prototypes into systems that hold up in the field.

Professional Experience

Research and Development Engineer · Contract **PT Aldzama** **Feb 2026 - Present**

Promoted from Demolition Robot Technician based on demonstrated engineering capability. Project specifics are confidential.

- Develop embedded control systems for industrial demolition robots, from schematic and PCB design through firmware
- Develop firmware in C/C++ utilizing microcontroller peripherals including UART, CAN bus, PWM, and digital I/O for real-time motor and actuator control
- Work on electro-hydraulic control modules, interfacing embedded control signals with proportional valve and actuator systems
- Perform testing, validation, and field troubleshooting of electronic and control systems at client sites
- Collaborate with cross-functional teams to transition prototypes into field-ready solutions

Demolition Robot Technician · Contract **PT Aldzama** **Sep 2025 - Jan 2026**

- Worked on maintenance, technical support, and troubleshooting for demolition robots
- Supported equipment improvement work on electrical and control systems
- Promoted to R&D Engineer role following a 3-month project that demonstrated engineering capability

Electronics Engineer · Internship **PT Venambak Kail Dipantara** **Dec 2024 - May 2025**

- Redesigned PCB layout with ESP32 for FeederVE control box, handling the full production cycle from schematic and PCB design to assembly and panel box integration for 50+ units
- Resolved electromagnetic interference (EMC/EMI) issues on the BoxVE system through PCB redesign, ensuring stable operation in outdoor environments
- Developed embedded software for AutoVE (autonomous mobile aerator) using Python on a low-cost mini PC

ROV Pilot and Engineer · Collaborative Project **PT Pertamina Hulu Energi West Madura Offshore** **Jul 2024 - Aug 2024**

- Led technical recovery of a non-operational ROV system: electrical diagnostics, control system restoration, and sea trials
- Served as on-site ROV pilot during offshore hull inspection survey
- Performed maintenance and troubleshooting on ROV control systems, cameras, and sensors

Maintenance and Improvement Engineer · Internship **PT Toshin Prima Fine Blanking (Indoprima Group)** **Feb 2024 - Jul 2024**

- Developed and deployed YOLOv4-based visual inspection system on Jetson Nano achieving ~29 FPS for real-time OK/NG quality control detection
- Built IoT monitoring system for 2 critical factory chiller units using Raspberry Pi with web dashboard, enabling real-time monitoring to prevent factory-wide downtime
- Performed routine maintenance and repair of industrial fine blanking machines

Penship EEPIS Marine Robotics Team **Mar 2022 - Nov 2024**

Embedded Developer Mar 2023 - Nov 2024

- Built entire electrical and embedded software systems from zero, progressing from Python+MAVLink to ROS-based autonomy stack
- Designed custom PCBs and developed computer vision pipelines (HSV to YOLO) for autonomous navigation on AUV and ASV platforms
- Contributed to team achievements in national underwater robotics competitions

General Manager Mar 2023 - Dec 2023

- Led 15-member team with 4 robot platforms, establishing weekly meeting culture from scratch
- Aligned hardware, software, and mechanical workstreams to competition deadlines

Hardware Specialist Mar 2022 - Feb 2023

- Engineered electrical systems for Autonomous Semi Submarine Vehicle: propulsion, power management, and waterproofing
- Designed custom PCBs aligned with mechanical specifications for underwater operation
- Supported software team with microcontroller programming for mission-based autonomous movements

Education

Bachelor of Applied Science, Electronics Engineering **Aug 2021 - Aug 2025**

Politeknik Elektronika Negeri Surabaya (PENS) Surabaya, Indonesia

- Final Project: “Jaladhi, Autonomous Surface Vehicle with Collision Avoidance”. Embedded system integration on Jetson Nano with ROS, computer vision, and multi-sensor fusion (camera and RPLIDAR). Awarded 2nd Place, PENS Final Project Competition

Machine Learning Cohort, Bangkit Academy **Google, Tokopedia, Gojek, Traveloka** **Aug 2024 - Dec 2024**

- Completed 5-month intensive program in Machine Learning, Deep Learning, TensorFlow, and Computer Vision

Awards and Recognitions

Most Active and Best Idea, Kaizen Event, PT Aldzama 2026
2nd Place, Final Project Competition, Applied Bachelor of Electronics Engineering, PENS 2025
National Finalist and Funded Proposal, Pekan Kreativitas Mahasiswa (PKM) Research & Technology, “Jalavahana” ASV Project, Ministry of Education 2024
3rd Place National, Underwater Robotics Competition (KRBAI), Kontes Robot Indonesia 2023
2nd Place Region 2, Underwater Robotics Competition (KRBAI), Kontes Robot Indonesia 2023

Certifications

Sea Survival and Offshore Emergency Procedure, PT Global Saftindo (Credential: GS-ADM-024-3594) Jul 2024 - Jul 2027

Technical Skills

Embedded Systems: STM32, ESP32, Arduino, AVR, microcontroller peripherals (Digital I/O, ADC, Timer, Counter, Watchdog, UART, SPI, I2C, CAN), real-time control firmware
PCB and Hardware Design: Schematic design and PCB layout (EasyEDA, Eagle, KiCAD), electronics component selection, power electronics, motor drivers, EMC/EMI troubleshooting, soldering and prototyping, oscilloscope and multimeter
Programming: C/C++, Python, JavaScript, Bash
Tools and Workflow: Git (version control), Linux, ROS, OpenCV, YOLO, TensorFlow, MQTT
Other: Electrical wiring and single-line diagram reading, testing and validation, field deployment and troubleshooting
Languages: Bahasa Indonesia (Native), English (Professional Working)